

Лабораторная работа № 12

Настройка службы времени NTP

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Цель:

Получить навыки синхронизации системного времени с помощью Chrony/NTP.



Исходное состояние времени

```
vagrant@client:~  
vagrant@server:~$ timedatectl  
    Local time: Sun 2026-09-06 11:40:47 UTC  
    Universal time: Sun 2026-09-06 11:40:47 UTC  
        RTC time: Sun 2026-09-06 11:40:47  
    Time zone: UTC (UTC, +0000)  
System clock synchronized: yes  
      NTP service: active  
    RTC in local TZ: no  
vagrant@server:~$ wffny  
bash: wffny: command not found...  
vagrant@server:~$ wffey  
bash: wffey: command not found...  
vagrant@server:~$ date  
Sun Sep  6 11:41:04 AM UTC 2026  
vagrant@server:~$ hwclock  
hwclock: Cannot access the Hardware Clock via any known method.  
hwclock: Use the --verbose option to see the details of our search for an access method.  
vagrant@server:~$ exit  
logout  
PS D:\work\ndbaranov\vagrant> vagrant ssh client  
==> client: The machine you're attempting to SSH into is configured to use  
==> client: password-based authentication. Vagrant can't script entering the  
==> client: password for you. If you're prompted for a password, please enter  
==> client: the same password you have configured in the Vagrantfile.  
vagrant@127.0.0.1's password:  
Last login: Sun Sep  6 11:16:34 2026 from 10.0.2.2  
vagrant@client:~$ timedatectl  
    Local time: Sun 2026-09-06 11:42:40 UTC  
    Universal time: Sun 2026-09-06 11:42:40 UTC  
        RTC time: Sun 2026-09-06 11:42:40  
    Time zone: UTC (UTC, +0000)  
System clock synchronized: yes  
      NTP service: active  
    RTC in local TZ: no  
vagrant@client:~$ dat  
bash: dat: command not found...  
vagrant@client:~$ ^C  
vagrant@client:~$ date  
Sun Sep  6 11:42:46 AM UTC 2026  
vagrant@client:~$ hwclock  
hwclock: Cannot access the Hardware Clock via any known method.  
hwclock: Use the --verbose option to see the details of our search for an access method.  
vagrant@client:~$
```

Смысл результата

Команды `timedatectl` и `date` показывают текущий часовой пояс, системное время и состояние синхронизации NTP на сервере.

Источники Chrony на сервере

```
root@server:~  
[root@server ~]# sudo -i  
dnf -y install chrony  
[root@server ~]# dnf -y install chrony  
Last metadata expiration check: 2:52:02 ago on Sun 06 Sep 2026 08:51:38 AM UTC.  
Package chrony-4.8-2.el10.x86_64 is already installed.  
Dependencies resolved.  
Nothing to do.  
Complete!  
[root@server ~]# chronyc sources  
MS Name/IP address         Stratum Poll Reach LastRx Last sample  
=====
```

^- as57164-151-0-2-53.htel.>	2	6	377	18	-5817us[+7405us]	+/-	24ms
^* mskm9-ntp01c.ntppool.yan>	2	6	377	9	+2789us[+17ms]	+/-	3411us
^- 31.186.134.105	2	6	377	27	-498us[+12ms]	+/-	58ms
^? 2a0d:d940:1c00::2	0	10	0	-	+0ns[+0ns]	+/-	0ns

```
[root@server ~]#
```

Смысл результата

chronyc sources выводит доступные внешние серверы времени, их состояние и измеренное смещение. Выбранный источник отмечается символом ^*.

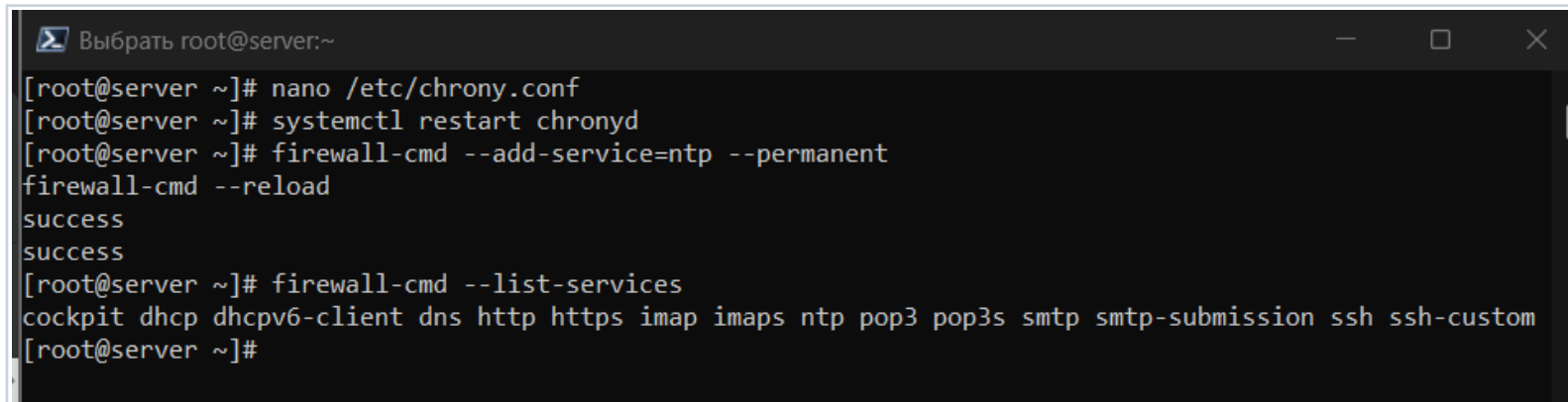
Конфигурация NTP-сервера

```
vagrant@client:~$ chronyc sources
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^- 193.106.93.116            2    6   377   16  -8151ns[ -88us] +/- 3979us
^- ntp2.mail.ru              2    6   277   24   -426us[ -426us] +/- 4046us
^* ntp1.vniiftri.ru          1    7   377   26   -427us[ -507us] +/- 3759us
^- mce.s-inform.net          2    8   377  150   +275us[ -85us] +/-  32ms
vagrant@client:~$
```

Смысл результата

В `chrony.conf` заданы внешние источники и разрешена выдача времени лабораторной подсети. Сервер может обслуживать `client` как локальный NTP-источник.

Открытие NTP в firewalld

A terminal window titled 'Выбрать root@server:~' with standard window controls. It contains the following commands and their outputs:

```
[root@server ~]# nano /etc/chrony.conf
[root@server ~]# systemctl restart chronyd
[root@server ~]# firewall-cmd --add-service=ntp --permanent
firewall-cmd --reload
success
success
[root@server ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps ntp pop3 pop3s smtp smtp-submission ssh ssh-custom
[root@server ~]#
```

Смысл результата

В межсетевом экране разрешён сервис ntp. После перезапуска chronyd сервер доступен по UDP-порту 123.

Настройка клиента Chrony

```
root@client:~$ sudo -i
root@client:~# nano /etc/chrony.conf
root@client:~# systemctl restart chronyd
root@client:~# chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
=====
^? spb-ntp01c.ntppool.yande> 0 6 0 - +0ns[ +0ns] +/- 0ns
^+ 89.188.118.150           4 6 7 2 +5824us[-3225us] +/- 100ms
^? scan-76.skipa.cyberok.ru 0 6 0 - +0ns[ +0ns] +/- 0ns
^? 2a02:6bf:f000:1:1::2     0 6 0 - +0ns[ +0ns] +/- 0ns
^? 45.141.102.99            2 6 1 2 +8477us[-573us] +/- 49ms
^+ time100.stupi.se         1 6 17 1 +5472us[+5472us] +/- 12ms
^? 2a02:6bf:f000:1:1::4     0 6 0 - +0ns[ +0ns] +/- 0ns
^? ns1.ooonet.ru            0 6 0 - +0ns[ +0ns] +/- 0ns
^- server.ndbaranov.net     2 6 17 3 -13ms[-8096us] +/- 95ms
root@client:~# chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
=====
^? spb-ntp01c.ntppool.yande> 0 7 0 - +0ns[ +0ns] +/- 0ns
^- 89.188.118.150           4 6 17 39 -1111us[+3574us] +/- 101ms
^? scan-76.skipa.cyberok.ru 0 7 0 - +0ns[ +0ns] +/- 0ns
^? 2a02:6bf:f000:1:1::2     0 7 0 - +0ns[ +0ns] +/- 0ns
^- 45.141.102.99            2 6 17 34 -40ms[-40ms] +/- 88ms
^* time100.stupi.se         1 6 17 39 +787us[+5472us] +/- 12ms
^? 2a02:6bf:f000:1:1::4     0 7 0 - +0ns[ +0ns] +/- 0ns
^? ns1.ooonet.ru            0 7 0 - +0ns[ +0ns] +/- 0ns
^- dhcp.ndbaranov.net.1.168> 2 6 17 38 -13ms[-8096us] +/- 95ms
root@client:~#
```

Смысл результата

На client в качестве предпочтительного источника указан server.ndbaranov.net. Старые внешние источники не мешают контролируемой проверке локального сервера.

Проверка источника клиента

```
root@client:~  
root@client:~# chronyc tracking  
reference ID      : C0248F82 (time100.stupi.se)  
stratum          : 2  
ref time (UTC)   : Sun Sep 06 11:47:15 2026  
system time      : 0.000000001 seconds fast of NTP time  
last offset      : +0.004684962 seconds  
MS offset        : 0.004684962 seconds  
frequency        : 535.234 ppm fast  
residual freq    : -597.912 ppm  
skew             : 1.452 ppm  
boot delay       : 0.023864377 seconds  
boot dispersion  : 0.038845364 seconds  
update interval  : 2.0 seconds  
leap status      : Normal  
root@client:~# chronyc sourcestats  
=====
```

Name/IP Address	NP	NR	Span	Frequency	Freq Skew	Offset	Std Dev
pb-ntp01c.ntppool.yande>	0	0	0	+0.000	2000.000	+0ns	4000ms
9.188.118.150	5	3	71	-47.045	252.681	-526us	1083us
can-76.skipa.cyberok.ru	0	0	0	+0.000	2000.000	+0ns	4000ms
a02:6bf:f000:1:1::2	0	0	0	+0.000	2000.000	+0ns	4000ms
5.141.102.99	4	4	6	-620.759	13356.681	-40ms	1197us
time100.stupi.se	5	4	71	-39.626	268.697	-139us	1101us
a02:6bf:f000:1:1::4	0	0	0	+0.000	2000.000	+0ns	4000ms
s1.ooonet.ru	0	0	0	+0.000	2000.000	+0ns	4000ms
ail.ndbaranov.net.1.168>	5	3	71	+42.352	597.224	-8664us	1303us

```
root@client:~#
```

Смысл результата

chronyc sources на клиенте показывает лабораторный server среди активных источников. Значения reach подтверждают регулярный обмен пакетами.

Статистика синхронизации

```
root@server:~  
[root@server ~]# chronyc sources  
MS Name/IP address         Stratum Poll Reach LastRx Last sample  
=====
```

^? 2a02:6bf:f000:1:1::2	0	8	0	-	+0ns[+0ns]	+/-	0ns
^* time100.stupi.se	1	6	77	32	+10ms[+15ms]	+/-	14ms
^? ns1.oonet.ru	0	8	0	-	+0ns[+0ns]	+/-	0ns
^- 89.188.118.150	4	6	157	32	+7736us[+7736us]	+/-	73ms
^? 2a02:6bf:f000:1:1::4	0	8	0	-	+0ns[+0ns]	+/-	0ns
^? scan-76.skipa.cyberok.ru	0	8	0	-	+0ns[+0ns]	+/-	0ns
^? 45.141.102.99	2	6	1	24	+13ms[+13ms]	+/-	52ms
^? 51.250.68.198	0	8	0	-	+0ns[+0ns]	+/-	0ns

```
[root@server ~]# chronyc tracking  
Reference ID      : C0248F82 (time100.stupi.se)  
Stratum           : 2  
Ref time (UTC)    : Sun Sep 06 11:48:08 2026  
System time       : 0.003752601 seconds fast of NTP time  
Last offset       : +0.004442720 seconds  
RMS offset        : 0.039470278 seconds  
Frequency         : 535.269 ppm fast  
Residual freq     : -295.317 ppm  
Skew              : 50.667 ppm  
Root delay        : 0.028414872 seconds  
Root dispersion   : 0.025130402 seconds  
Update interval   : 64.2 seconds  
Leap status       : Normal  
[root@server ~]#
```

Смысл результата

Команды `tracking` и `sourcestats` выводят смещение часов, частотную поправку и качество измерений. Малое смещение означает устойчивую синхронизацию.

Provision Chrony

```
PowerShell 7 (x64)
PS D:\work\ndbaranov\vagrant> vagrant validate
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
Vagrantfile validated successfully.
PS D:\work\ndbaranov\vagrant> vagrant provision server --provision-with "server ntp"
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Running provisioner: server ntp (shell)...
server: Running: C:/Users/barash/AppData/Local/Temp/vagrant-shell20260906-11372-sg9rmc.sh
server: Provisioning NTP server
server: Install needed packages
server: Last metadata expiration check: 2:58:52 ago on Sun 06 Sep 2026 08:51:38 AM UTC.
server: Package chrony-4.8-2.el10.x86_64 is already installed.
server: Dependencies resolved.
server: Nothing to do.
server: Complete!
server: Configure chrony
server: Configure firewall
server: Warning: ALREADY_ENABLED: ntp
server: success
server: success
server: Restart chronyd service
server: NTP server provisioning completed
PS D:\work\ndbaranov\vagrant> vagrant provision client --provision-with "client ntp"
==> client: Running provisioner: client ntp (shell)...
client: Running: C:/Users/barash/AppData/Local/Temp/vagrant-shell20260906-7644-ch2z4f.sh
client: Provisioning NTP client
client: Install needed packages
client: Last metadata expiration check: 1:11:29 ago on Sun 06 Sep 2026 10:39:19 AM UTC.
client: Package chrony-4.8-2.el10.x86_64 is already installed.
client: Dependencies resolved.
client: Nothing to do.
client: Complete!
client: Configure chrony
client: Relabeled /etc/lvm/devices/system.devices from system_u:object_r:lvm_metadata_t:s0 to system_u:object_r:lvm_etc_t:s0
client: Relabeled /etc/lvm/devices/backup/system.devices-20260906.095343.0005 from system_u:object_r:lvm_metadata_t:s0 to system_u:object_r:lvm_etc_t:s0
client: Relabeled /etc/foomatic/hashtes.d/hashtes.new from system_u:object_r:etc_runtime_t:s0 to system_u:object_r:etc_t:s0
client: Restart chronyd service
client: NTP client provisioning completed
PS D:\work\ndbaranov\vagrant>
```

Смысл результата

Сценарии server-ntp и client-ntp повторно применяют роли сервера и клиента. Финальный вывод подтверждает запуск chronyd на обоих узлах.

Результат

Chrony синхронизирует часы сервера с внешними источниками и раздает время клиенту.
Команды `sources`, `tracking` и `sourcestats` подтверждают выбор источника и величину поправки.